

几种可积高阶方程

几个特殊的二阶方程.

1.

$$y'' = f(x), \quad y(x) = \int_0^x \int_0^x f(t) dt dx = \int_0^x (x-t)f(t) dt + c_1 x + c_2.$$

2.

$$y'' = f(y'),$$

令

$$z = y', \quad z' = f(z), \quad \frac{dz}{f(z)} = dx, \quad x = \int_{z_0}^z \frac{dt}{f(t)} + c.$$

3.

$$y'' = f(x, y'),$$

令 $z = y'$,

$$z' = f(x, z).$$

这是一阶常微分方程.

4.

$$y'' = f(y),$$

可得

$$y'' y' = \frac{d}{dx} \left(\frac{1}{2} (y')^2 \right) = f(y) y' = f(y) \frac{dy}{dx} = \frac{d(\int f(y) dy)}{dx}, \quad (y')^2 = 2 \int f(y) dy + c_1.$$

$$dx = \frac{dy}{\left(\sqrt{2 \int f(y) dy + c_1} \right)}, \quad x = \int \frac{dy}{\sqrt{2 \int_{y_0}^y f(t) dt + c_1}} + c_2.$$

5.

$$y'' = f(y, y'),$$

令 $z = y'$, $y'' = \frac{dy'}{dx} = \frac{dy'}{dy} \frac{dy}{dx} = \frac{dz}{dy} z$, 原方程变为

$$\frac{dz}{dy} z = f(y, z),$$

这是一阶常微分方程.

6.

$$y'' = f(ax + by), \quad b \neq 0, \quad a, b \text{ 是常数.}$$

令 $z = ax + by$, 可得:

$$z'' = bf(z),$$

这是4的形式.

$$x = \int \frac{dz}{\sqrt{2b \int_{z_0}^z f(t) dt + c_1}} + c_2.$$

几种可积的高阶方程.

I.

$$y^{(n)} = f(x),$$
$$y(x) = \int_0^x \cdots \int_0^x f(t) (dt)^n + \sum_{k=0}^n c_k x^k = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} f(t) dt + \sum_{k=0}^{n-1} c_k x^k.$$

II.

$$F(y^{(n-1)}, y^{(n)}) = 0.$$

若方程可就 $y^{(n)}$ 解出:

$$y^{(n)} = f(y^{(n-1)}),$$

则可令 $y^{(n-1)} = z$ 而化成

$$z' = f(z).$$

这是变量分离的方程, 解之得

$$\int \frac{dz}{f(z)} = x + c_1.$$

设从上式可解出 $z = \omega(x, c_1)$, 则

$$y^{(n-1)} = \omega(x, c_1).$$

由此可得到

$$y(x) = \frac{1}{(n-2)!} \int_0^x (x-t)^{n-2} \omega(t, c_1) dt + \sum_{k=0}^{n-2} c_k x^k.$$

$$\text{III. } F(y^{(n-2)}, y^{(n)}) = 0.$$

设方程可就 $y^{(n)}$ 解出:

$$y^{(n)} = f(y^{(n-2)}),$$

则令 $z = y^{(n-2)}$ 而将它化成

$$z'' = f(z).$$

上式两端乘以 $2z'$ 得

$$2z'z'' = 2f(z)z',$$

即

$$\begin{aligned} \frac{d(z')^2}{dx} &= 2z'f(z), \quad d(z')^2 = 2f(z)dz, \\ (z')^2 &= 2 \int f(z)dz + c_1, \end{aligned}$$

由此可得

$$z' = \sqrt{2 \int f(z)dz + c_1}.$$

用分离变量法求得 $z = \omega(x, c_1, c_2)$, 则

$$y^{(n-2)} = \omega(x, c_1, c_2).$$

这是类型I的方程, 积分 $n-2$ 次即可求得原方程的通解.

例.

$$a^2 \frac{d^4 y}{dx^4} = \frac{d^2 y}{dx^2}.$$

令 $y'' = z$, 则

$$a^2 z'' = z.$$

以 $2z'dx$ 乘以上式两端,

$$a^2 d(z')^2 = 2zdz, \quad a^2 (z')^2 = z^2 + c_1, \quad az' = \sqrt{z^2 + c_1}.$$

用分离变量法:

$$\frac{dz}{\sqrt{z^2 + c_1}} = \frac{dx}{a}, \quad \ln(z + \sqrt{z^2 + c_1}) = \frac{x}{a} + \ln c_2,$$

由此可以解出

$$z + \sqrt{z^2 + c_1} = c_2 e^{\frac{x}{a}}. \quad (0.1)$$

为了解出 z , 将方程(0.1)写成

$$\frac{1}{z + \sqrt{z^2 + c_1}} = \frac{1}{c_2} e^{-\frac{x}{a}}.$$

将上式左端有理化, 然后两端乘以 $-c_1$, 得到

$$z - \sqrt{z^2 + c_1} = -\frac{c_1}{c_2} e^{-\frac{x}{a}}. \quad (0.2)$$

将等式(0.1)和等式(0.2)相加, 除以2得

$$z = \frac{c_2}{2} e^{\frac{x}{a}} - \frac{c_1}{2c_2} e^{-\frac{x}{a}}.$$

用 y'' 代替 z ,

$$y'' = \frac{c_2}{2} e^{\frac{x}{a}} - \frac{c_1}{2c_2} e^{-\frac{x}{a}}.$$

对 x 积分两次得

$$y = \frac{a^2 c_2}{2} e^{\frac{x}{a}} - \frac{a^2 c_1}{2c_2} e^{-\frac{x}{a}} + c_3 x + c_4 = A e^{\frac{x}{a}} + B e^{-\frac{x}{a}} + C x + D,$$

这里 A, B, C, D 都是任意常数.